Neural Network Task 2

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Neural Network Task 2 Report

Multi-Layer Perceptron with Backpropagation

Bird Species Classification (Penguins Dataset)

1. Overview

This report presents the implementation of a Multi-Layer Perceptron (MLP) trained with Backpropagation to classify bird species into three classes: Adelie, Chinstrap, and Gentoo.

2. Dataset

- 5 input features per sample
- 3 classes (Adelie, Chinstrap, Gentoo), 50 samples each
- Training: first 30 samples per class (90 total)
- Testing: remaining 20 samples per class (60 total)

3. Network Setup

- Input layer: 5 neurons (one per feature)
- Hidden layers: configurable via UI
- Output layer: 3 neurons (one per class)
- Weight initialization: Xavier/Glorot uniform
- Bias: optional (toggled in the UI)
- Activation functions tested: Sigmoid, Hyperbolic Tangent (tanh)

4. Best Results

The table below shows the best accuracy achieved for each activation function along with the parameters used.

Activation	Train Acc	Test Acc	LR	Epochs	#Layers	Hidden Nodes
Sigmoid	98.9%	96.7%	0.05	200	1	[4]
Tanh	100%	98.3%	0.05	200	2	[4, 4]

5. Screenshots

5.1 Sigmoid — Best Run

5.2 Tanh — Best Run

6. Observations

- Both activation functions were able to separate the three bird classes.
- Tanh generally converges faster than Sigmoid due to its zero-centered output.
- Sigmoid can achieve similar accuracy but may require more epochs or a higher learning rate.

- Adding bias improved accuracy in most configurations.